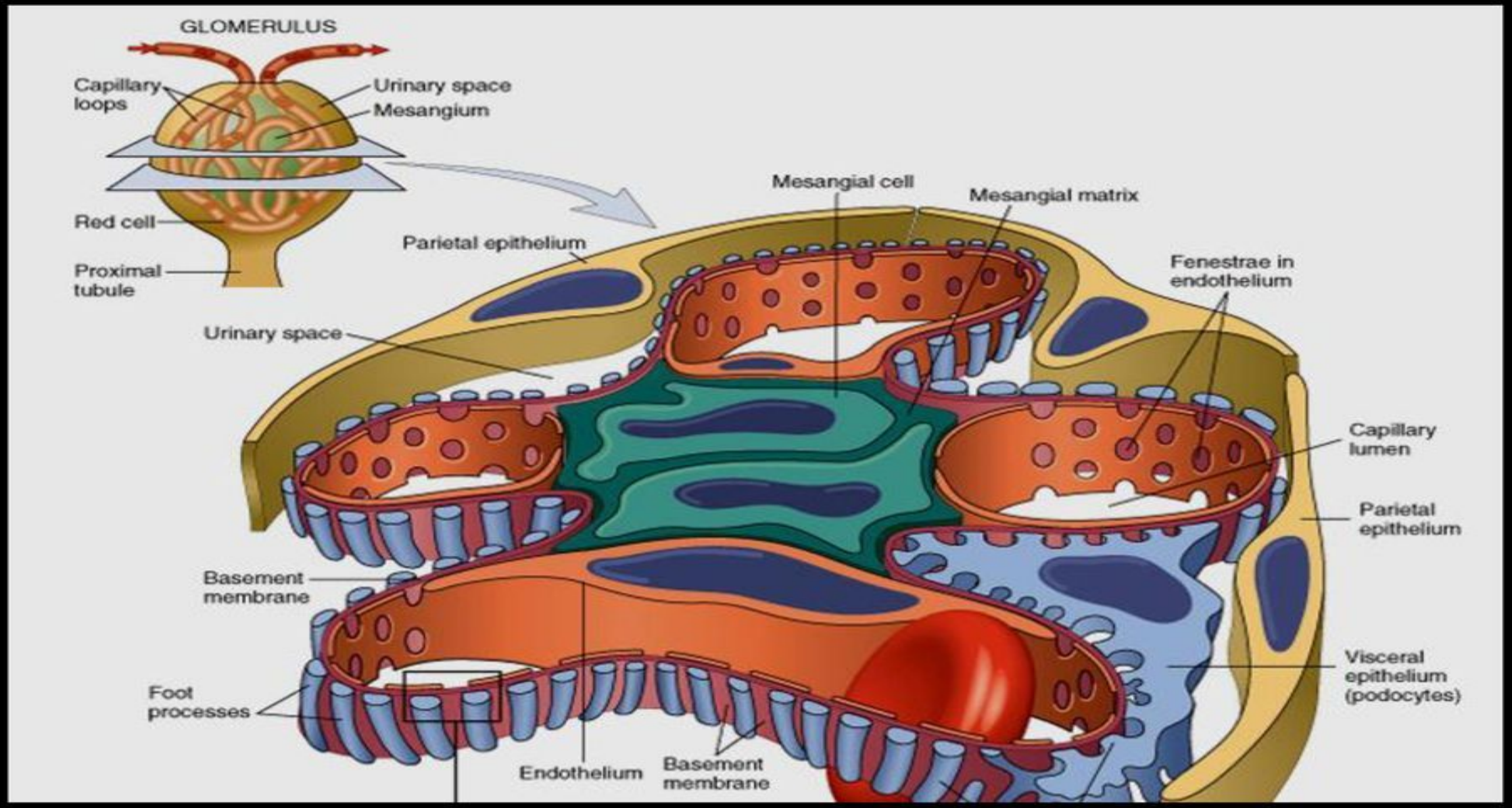
Glomerulus

- A tuft of capillaries situated within a Bowman's capsule at the end of a renal tubule in the vertebrate kidney that filters waste products from the blood and thus initiates urine formation.
- The basic filtration unit of the kidney.

Schematic diagram of a lobe of a normal glomerulus.



GLOMERULAR DISEASES

DEFINITION AND CLASSIFICATION

- Glomerulonephritis (GN) or Bright's disease is the term used for diseases that primarily involve the renal glomeruli.

- classify - 2 broad groups:

- I. **Primary glomerulonephritis** in which the glomeruli are the predominant site of involvement.

- II. **Secondary glomerular diseases** include certain systemic and hereditary diseases which secondarily affect the glomeruli

classification is based on histology. Subdivided:

a) diffuse (all glomeruli)
b) global (entire glomerulus)
c) focal (portion of glomeruli)
d) segmental (part of each glomerulus)
e) mesangial (affecting mesangial region)

www.freelivedoctor.com

PATHOGENESIS

1. **Circulating immune complex deposition:**

Antigen + Antibody – deposition in glomeruli – binding with complement – inflammation – glomerular injury.

- Antigen may be exogenous or endogenous

2. **Antibodies directed against antigen on glomerular capillary membrane**:

- Anti – GBM antibody disease – antigen fixed in the GBM e.g good pasture syndrome
- Antibodies against non GBM antigen

NEPHROTIC SYNDROME

Clinical complex characterized by

- Heavy proteinuria (3.5 g/day)
- Hypoalbuminemia
- Edema
- Hypelipidemia
- Hyperlipiduria

Plasma protein <3gm

Increase lipoprotein synthesis in liver

Proteinuria:

- daily loss of protein 3.5 gm or more of protein
- Injury to the capillary wall of the glomeruli result — increase permeability to the plasma protein — allow to — escape from plasma

CAUSES OF NEPHROTIC SYNDROME

I. PRIMARY GLOMERULONEPHRITIS
1. Minimal change disease *(most common in children)*
2. Membranous GN *(most common in adults)*
3. Membranoproliferative GN
4. Focal segmental glomerulosclerosis
5. Focal and diffuse proliferative GN
6. IgA nephropathy

II. SYSTEMIC DISEASES
1. Diabetes mellitus
2. Amyloidosis
3. SLE

III. SYSTEMIC INFECTIONS
1. Viral infections (HBV, HCV, HIV)
2. Bacterial infections (bacterial endocarditis, syphilis, leprosy)
3. Protozoa and parasites (*P. falciparum* malaria, filariasis)

CAUSES OF NEPHROTIC SYNDROME

IV. HYPERSENSITIVITY REACTIONS

1. Drugs (heavy metal compounds like gold and mercury, other drugs like penicillamine, trimethadione and tolbutamide, heroin addiction)
2. Bee stings, snake bite, poison ivy

V. MALIGNANCY

1. Carcinomas
2. Myeloma
3. Hodgkin's disease

VI. PREGNANCY

Toxaemia of pregnancy

VII. CIRCULATORY DISTURBANCES

1. Renal vein thrombosis
2. Constrictive pericarditis

VIII. HEREDITARY DISEASES

1. Alport's disease
2. Fabry's disease
3. Nail-patella syndrome

Minimal change disease (Lipoid Nephrosis):

- Most frequent cause of NS in children.
- Peak incidence is between 2 and 6 years of age.
- Follows a respiratory infection or prophylactic immunization.
- Proteinuria is highly selective (albumin).
- Dramatic response to corticosteroid therapy.
- Tubular reabsorption of lipoproteins passing through diseased glomeruli (lipoid nephrosis).

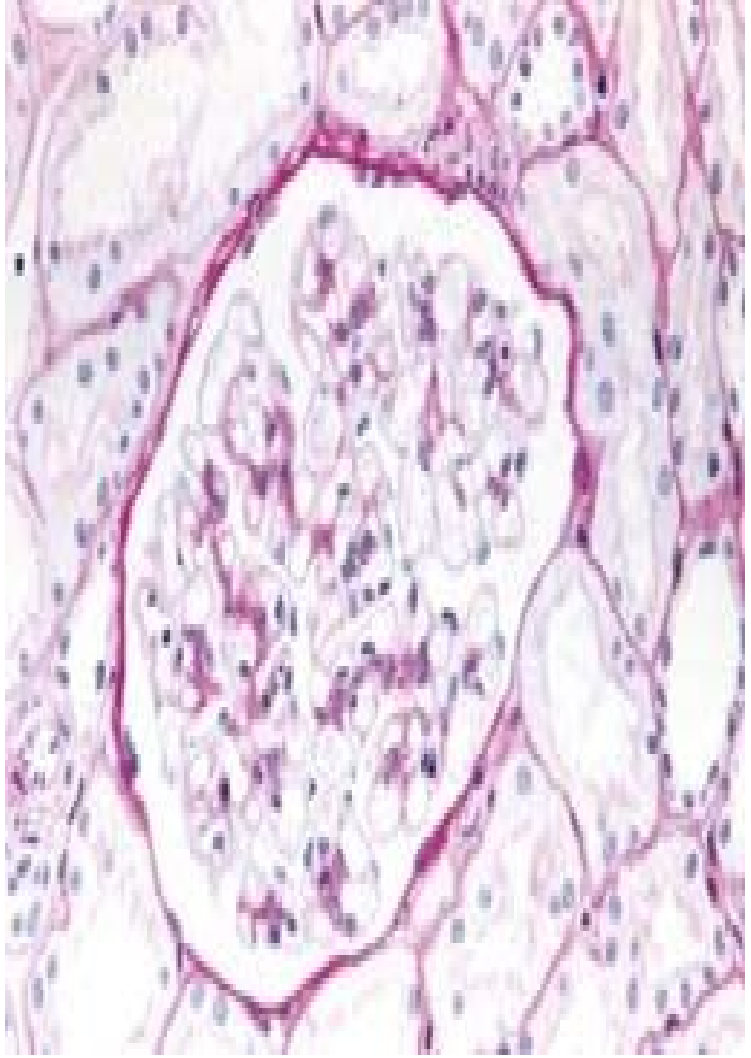
Morphology:

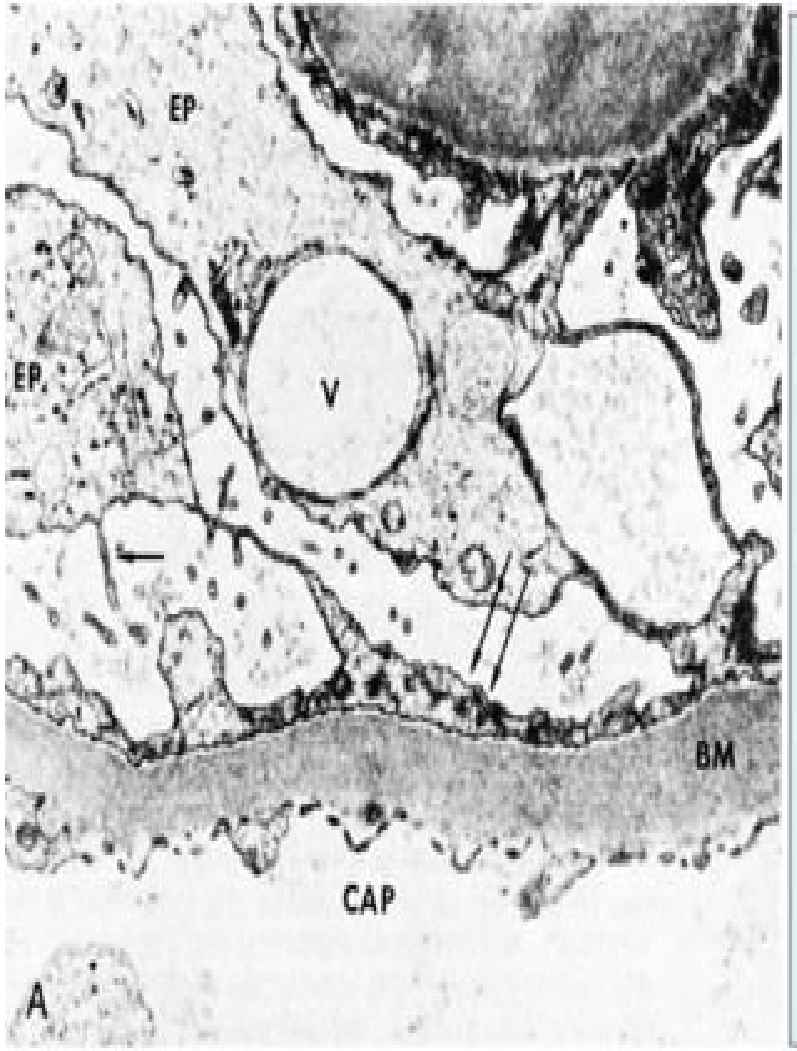
- Light microscopy: glomeruli are normal.
- Immunofluorescence: no immunoglobulin or complement deposits.
- Electron microscopy:

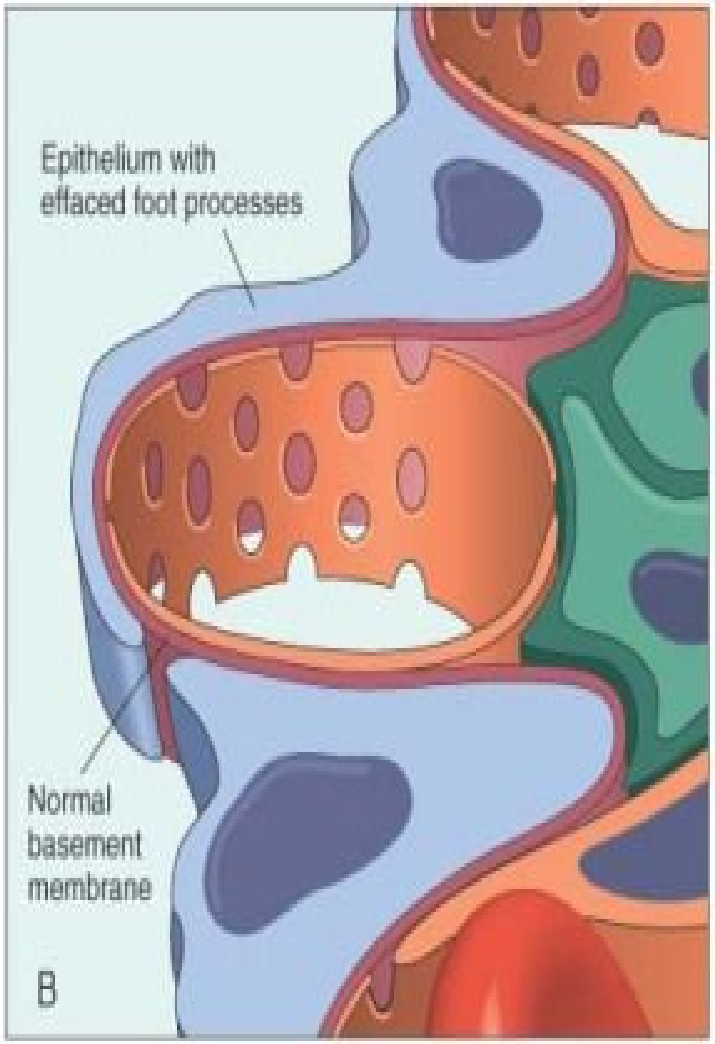
➢ GBM appears normal.

➢ No electron-dense material is deposited.

➢ The visceral epithelial cells show diffuse effacement of foot processes which are replaced by a rim of cytoplasm which shows vacuolization and hyperplasia of villi.







MCD: Light microscopy, Glomerulus stained

with PAS: Normal basement membrane and absence of proliferation.

A, Ultrastructural characteristics of minimal change disease: effacement of foot processes (double arrows), absence of deposits, vacuoles (V), and microvilli in visceral epithelial cells (single arrow). B, Schematic representation of minimal change disease, showing diffuse effacement of foot processes

Membranous glomerulopathy (MG):

- The most common cause of NS in adults.
- Cause;
- **Primary MG** (85% of cases):- idiopathic (renal auto antigen).
- **Secondary MG:-**

✓Drugs (penicillamine)

✓Malignancy (Ca. of lung, colon and melanoma)

✓SLE ;

✓Infections (chronic hepatitis B or C, malaria, syphilis)

✓Autoimmune thyroiditis.

Pathogenesis:

- Immune complex-mediated disease.
- In contrast to minimal change disease, the proteinuria is nonselective (albumin and globulin are both excreted.
- Does not respond well to corticosteroid therapy.

Morphology:

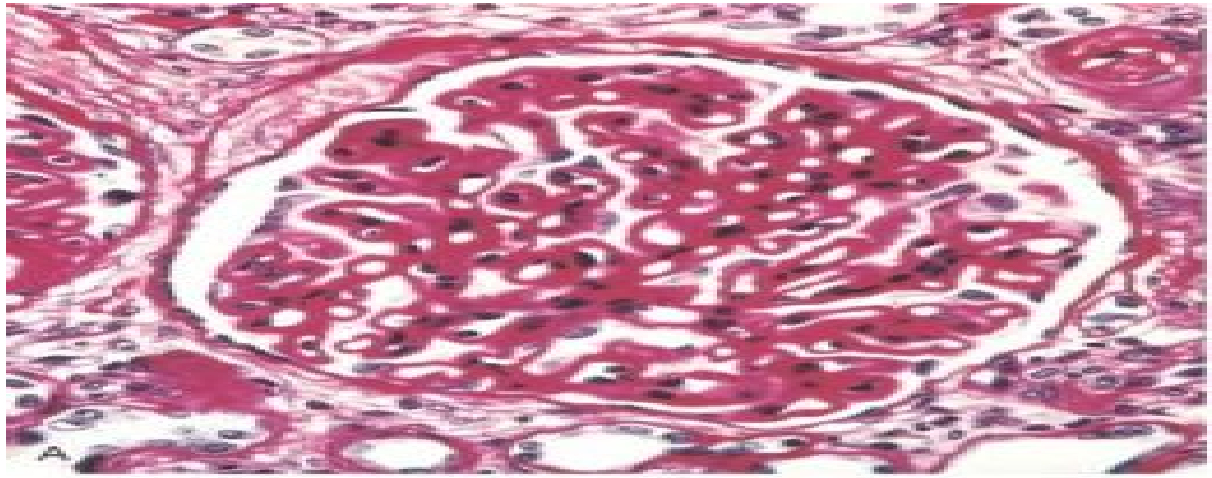
Feature of light microscope

- Diffuse thickening of glomerular basement membrane

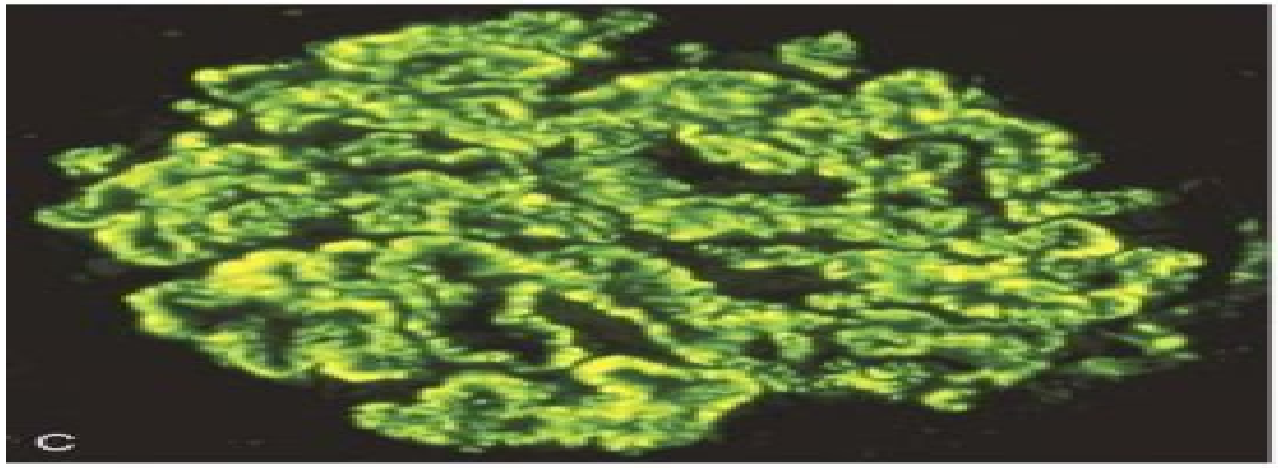
Feature of electron microscope:

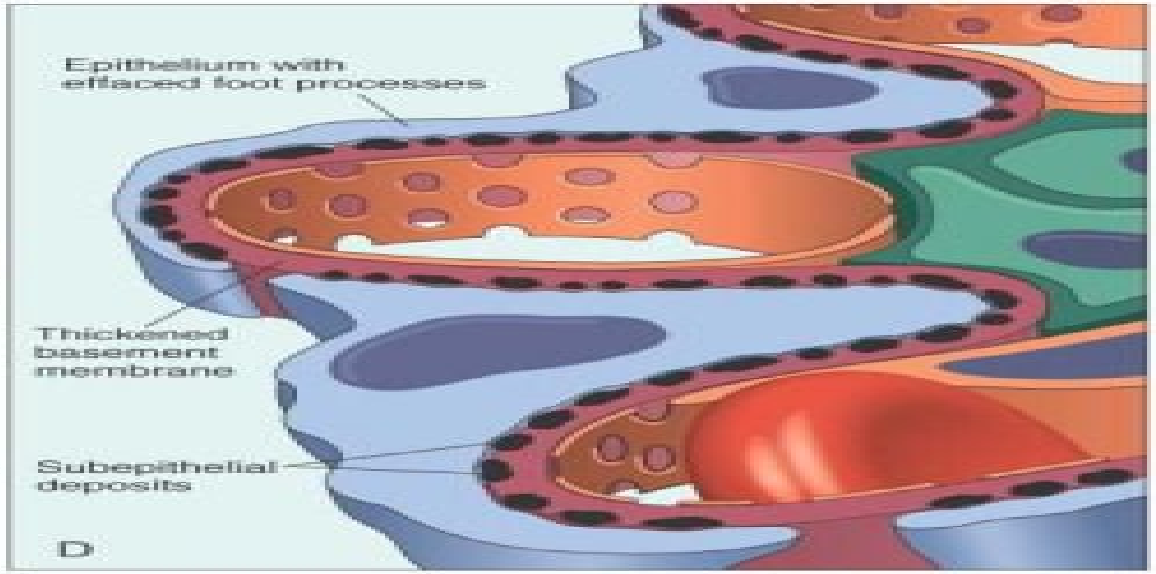
- Apparent thickening of GBM is caused by subepitheial deposits that are separated from each other by small spike in GBM matrix.

A **: PAS stain, diffuse thickening of capillary wall without increase in number of cells.** ***B*** **: Electron micrograph showing electron-dense deposits (*arrow*) along the epithelial side of GBM (*B*). Note the obliteration of foot process overlying the deposits… capillary lumen (CL); endothelium (End); epithelium (Ep).** ***C*** **: granular immunofluorescent deposits of IgG along GBM.** ***D*** **: Diagrammatic representation.**









Focal Segmental Glomerulosclerosis (FSGS):

- Sclerosis of some glomeruli (focal); and in affected glomeruli only a portion is involved (segmental).
- Sclerosis within capillary tufts of deep juxtamedullary glomeruli.

❖ Clinically: nephrotic syndrome.

Causes:

✓Genetic

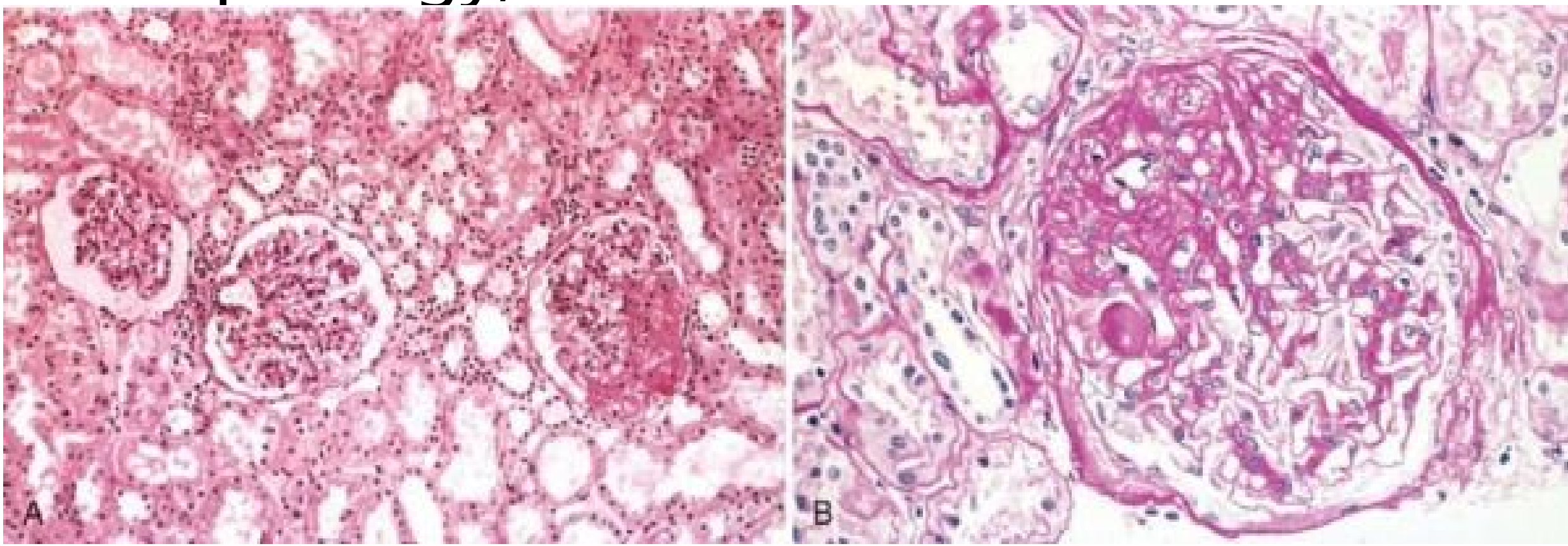
✓drugs (anabolic steroids),toxins (heroin)

✓Infection; HIV

✓chronic pyelonephritis, diabetes, and hypertension.

✓ idiopathic.

Morphology;



Focal segmental glomerulosclerosis, PAS stain. A: Low-power view showing segmental sclerosis in one of three glomeruli (at 3 o'clock). B: High-power view showing hyaline in sudation and lipid (small vacuoles) in sclerotic area.

Membranoproliferative glomerulonephritis
MPGN

Classification based on Immunofluorescence

- Immunoglobulin mediated (monoclonal or polyclonal) or complement mediated

✓ Immune complex/ monoclonal immunoglobulin mediated MPGN(activation of classic complement pathway)

✓ C3 glomerulopathies

Protein or hematuria

MPGN

- Primary; idiopathic
- Secondary MPGN arises in:- SLE, Hepatitis B or C infection, α1-Antitrypsin deficiency, and Malignant diseases (CLL, lymphoma).

Previous classification;

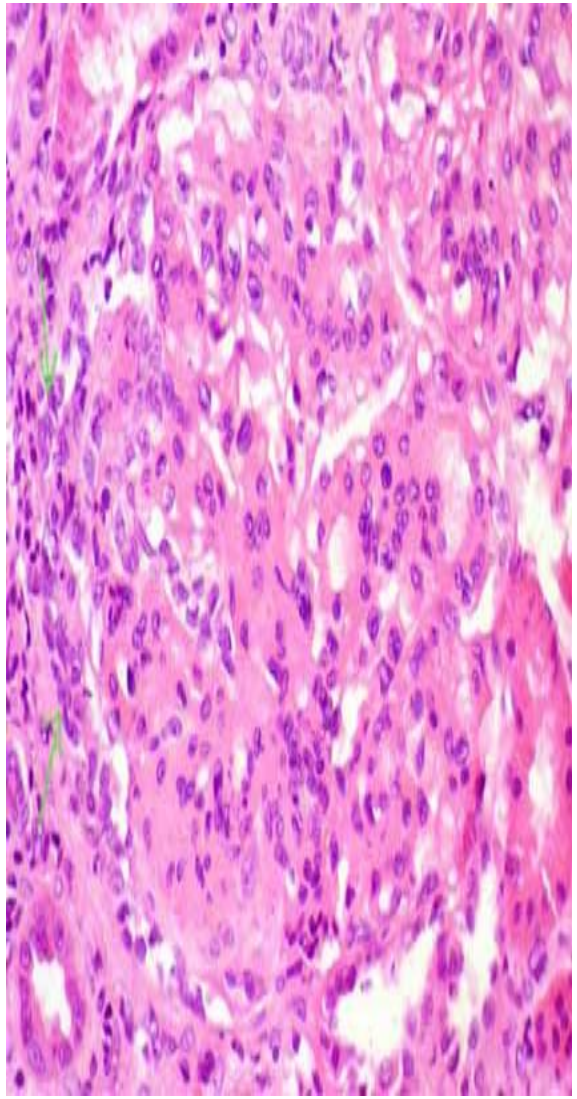
- Primary MPGN is subdivided into two types on basis of ultrastructural, immunofluorescent, and pathologic findings:- type I and type II MPGN (dense-deposit disease).

Morphology:

- *Light microscopy:*

- The glomeruli are large, lobular, and hypercellular.
- The hypercellularity is due to proliferation of cells in mesangium, and endocapillary cell proliferation (capillary endothelium and infiltrating leukocytes).
- Increased mesangial matrix.
- The GBM is thickened, often focally.
- The glomerular capillary wall shows a "double-contour" or "tram-track" appearance caused by "duplication" of GBM.

Characterized by endocapillary and mesangial hypercellularity, mesangial and subendothelial deposits and duplicating of glomerular basement membrane.





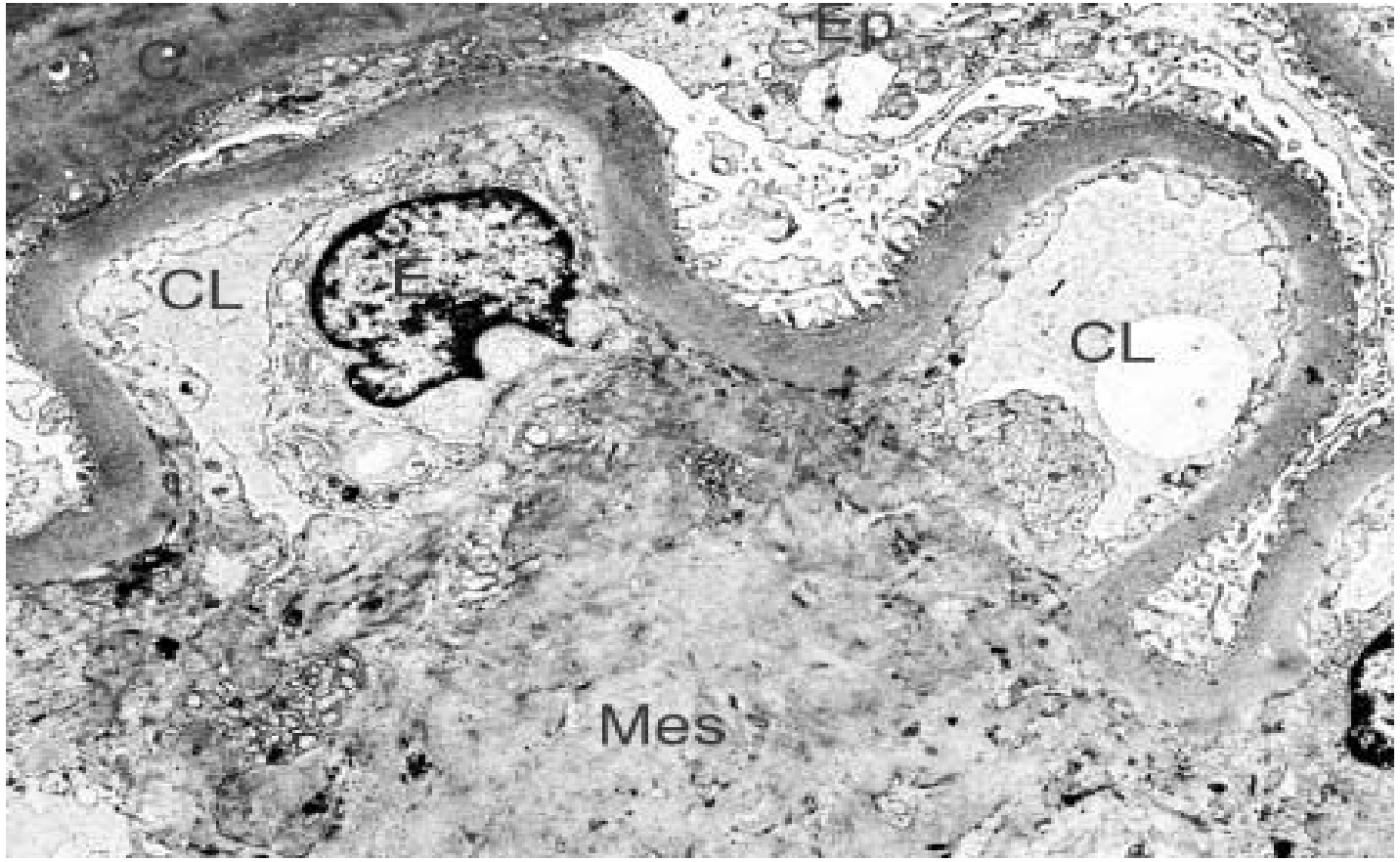
Duplication of GBM with double contour / tram track appearance.

Diabetic nephropathy

- Clinically:

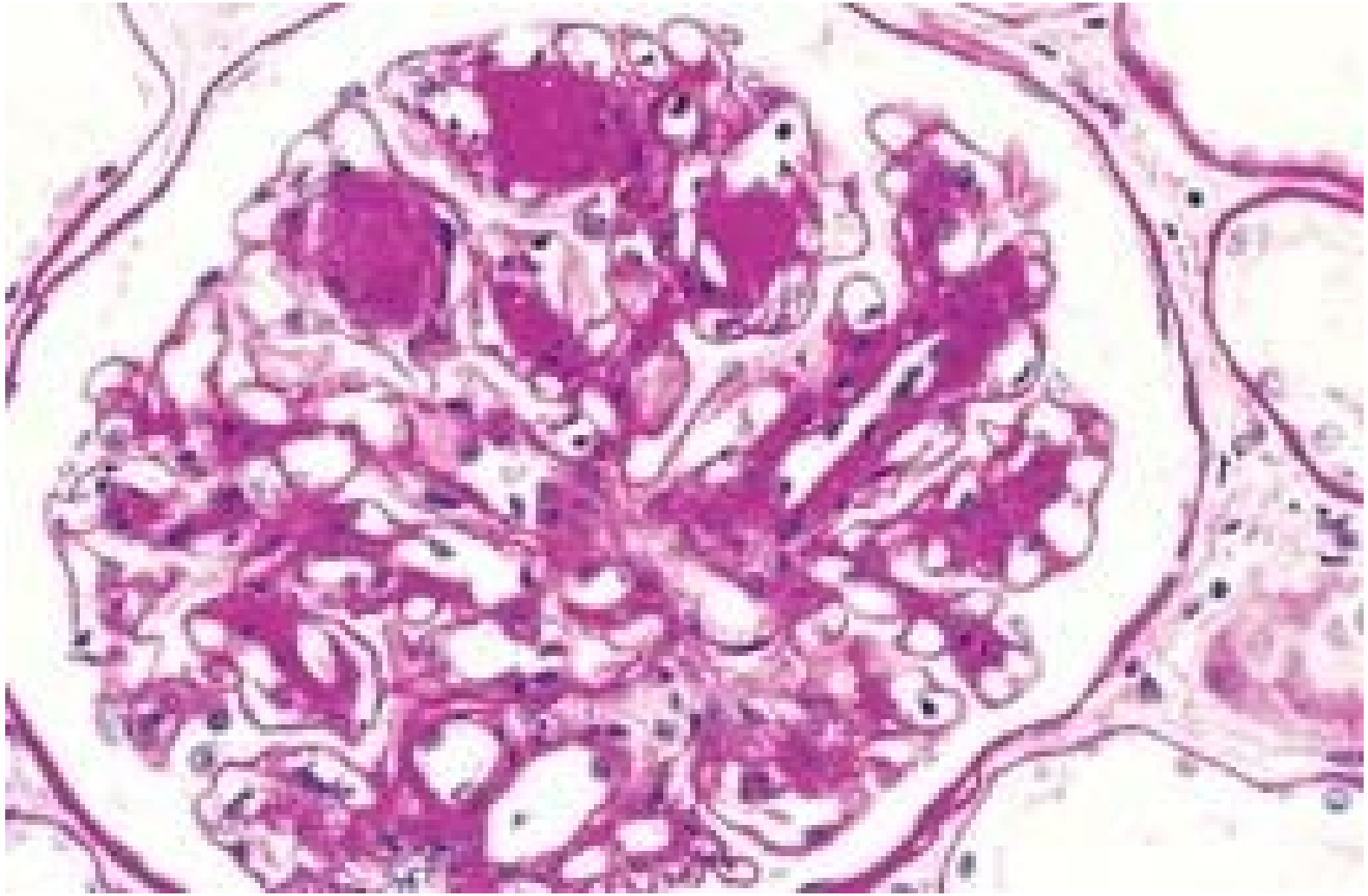
- Three glomerular syndromes:
  1. Non-nephrotic proteinuria: preceded by microalbuminuria.
  2. Nephrotic syndrome.
  3. Chronic renal failure.
- Diabetes also causing: hyalinizing arteriolar sclerosis, Increase susceptibility to acute pyelonephritis and papillary necrosis.

- Pathogenesis: Two processes play a role:
  - **Metabolic defect:** the advanced glycosylation end products lead to thickened GBM and increased mesangial matrix.
  - **Hemodynamic effects:** associated with glomerular hypertrophy which contributes to glomerulosclerosis.
- Morphology:
  - GBM thickening.
  - Increase in mesangial matrix result in:
    - Diffuse glomerulosclerosis.
    - Nodular glomerulosclerosis (Kimmelstiel-Wilson disease):- ovoid or spherical laminated PAS-positive nodules lie within the mesangial core surrounded by dilated capillary loops.



Electron micrograph: massive increase in mesangial matrix (Mes) encroaching on glomerular capillary lumen (CL). The GBM and Bowman capsule (C) are markedly thickened… Ep: epithelium; E: endothelium.

Nodular glomerulosclerosis (PAS stain): Note ovoid or spherical laminated PAS-positive nodules lie within the mesangial core surrounded by dilated capillary loops.



Renal amyloidosis:

- Another cause of nephrotic syndrome.
- Subendothelial and mesangial amyloid deposits.
- The amyloid can be identified by:-

o Special Amyloid stain (Congo red stain) gives:

➢ Red color under light microscope.

➢ Apple green birefringence under polarized light.

o Cross fibrillary pattern of amyloid by electron microscopy.

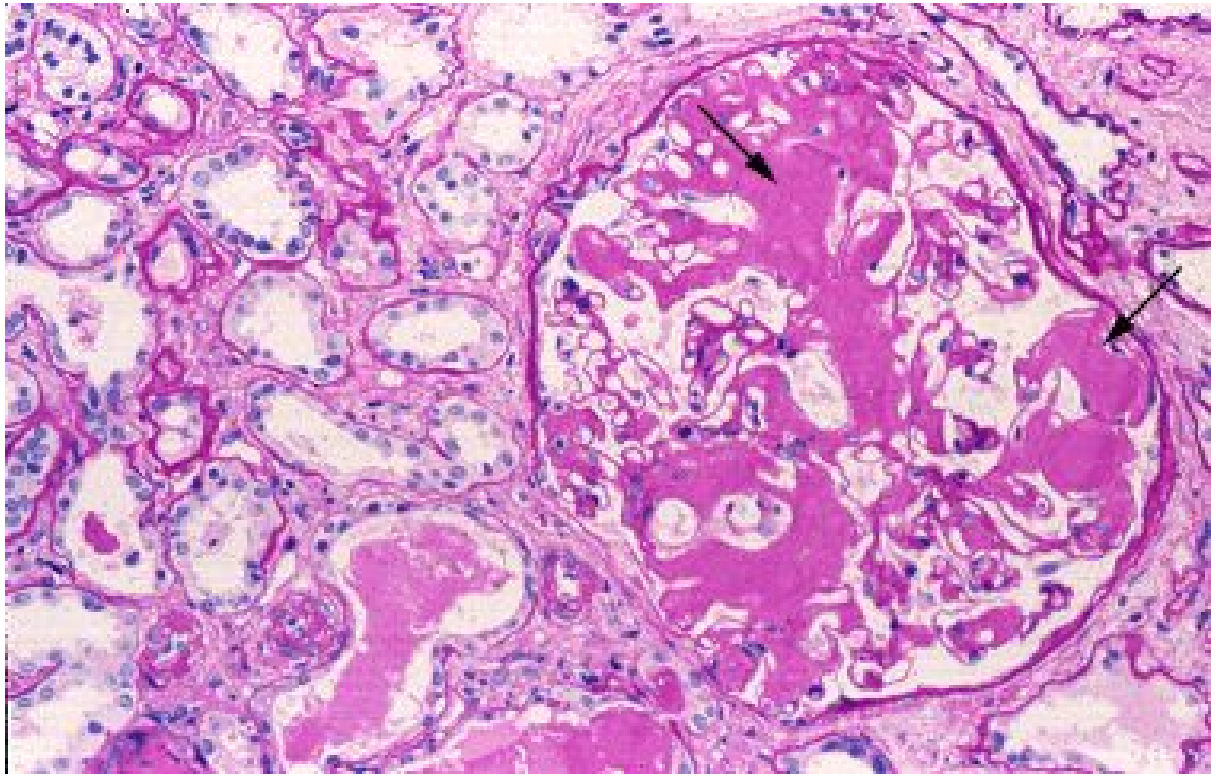
- There are two types:

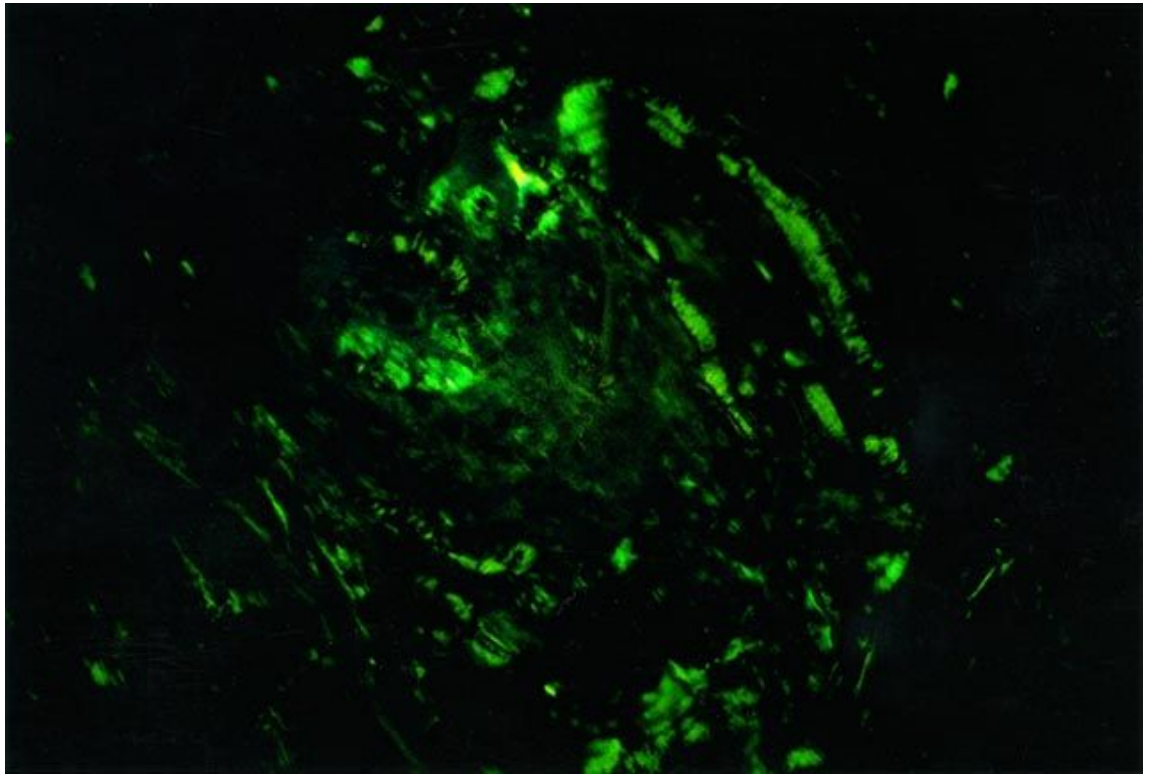
o Primary (AL) amyloidosis: in patients with plasma cell dyscrasias (Multiple myeloma) and consists of monoclonal light chains (kappa or lambda).

o Secondary (AA) amyloidosis: in patients with neoplasia or chronic inflammation (T.B) and is associated with serum amyloid A protein produced in liver.

There are two types:

- Primary (AL) amyloidosis: in patients with plasma cell dyscrasias (Multiple myeloma) and consists of monoclonal light chains (kappa or lambda).
- Secondary (AA) amyloidosis: in patients with neoplasia or chronic inflammation (T.B) and is associated with serum amyloid A protein produced in liver.





Lupus nephropathy:

- The severity of renal lesion often determines the prognosis.
- Classified by (WHO) into six renal patterns:

(1) Type I: No observable renal involvement.

(2) Type II: Mesangial proliferative form of lupus nephropathy.

(3) Type III: Focal proliferative form.

(4) Type IV: Diffuse proliferative form:- Characteristic change include wire-loop lesion (wire loop thickening of GBM resulting from subendothelial immune complex deposition).

(5) Type V: Membranous form (global or segmental subepithelial immune deposits).

(6) Type VI: Advanced sclerosis lupus nephritis.

Wire loop lesions

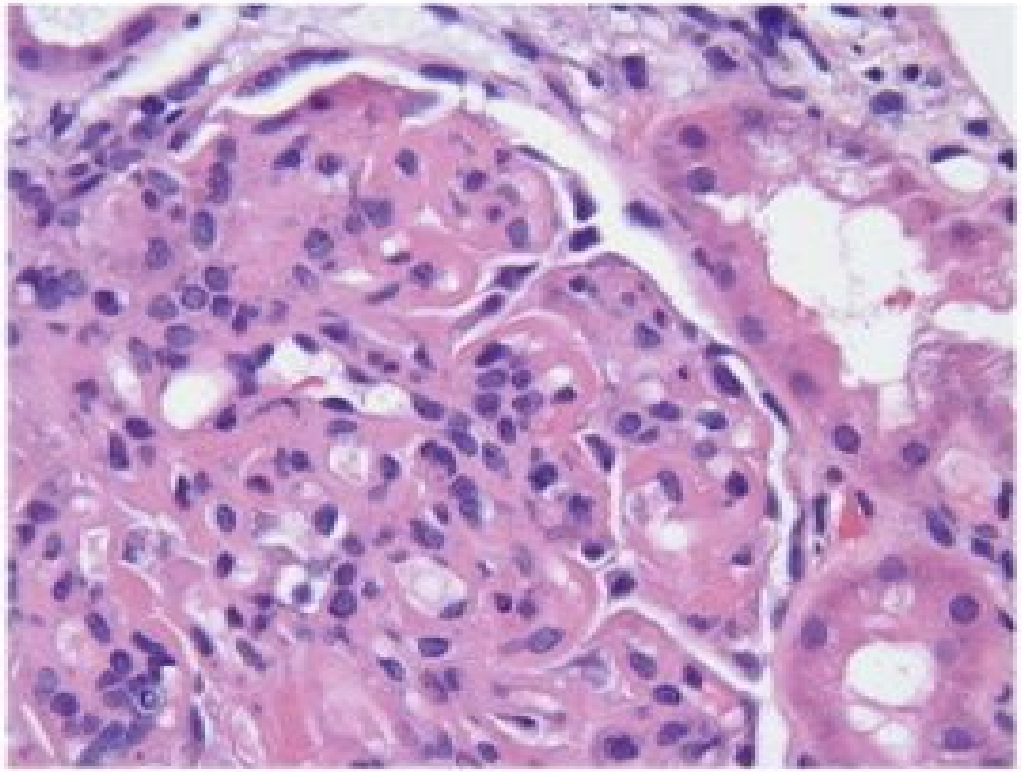


Figure 1 Wire loop. Glomerular capillary walls are thickened by wire-loop deposits showing rigid, band-like thickening of the glomerular basement membranes by intensely eosinophilic, glassy material corresponding to subendothelial deposits. (Haematoxylin and eosin stain).



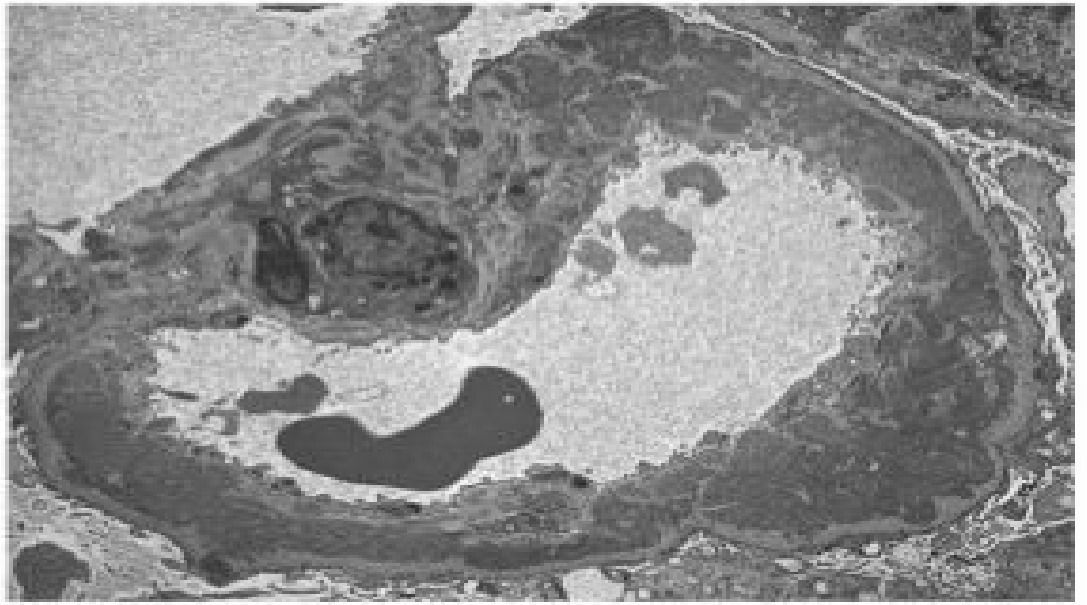


Figure 2 Wire loop. As illustrated in this electron micrograph, the wire loop represents large subendothelial electron-dense deposits composed of immune complexes.

Nephritic syndrome:

- Characterized by inflammation in the glomeruli.
- Clinical findings
  - Hematuria (red blood cells & red cell casts in urine)
  - proteinuria usually (subnephrotic) with or without edema.
  - Oliguria
  - Azotemia
  - Hypertension

Post streptococcal GN (Acute Proliferative GN):

- 1 to 4 weeks after a streptococcal infection of pharynx or skin (impetigo).
- Mostly in children (6 to 10 years), but adults can be affected.
- Etiology and Pathogenesis:

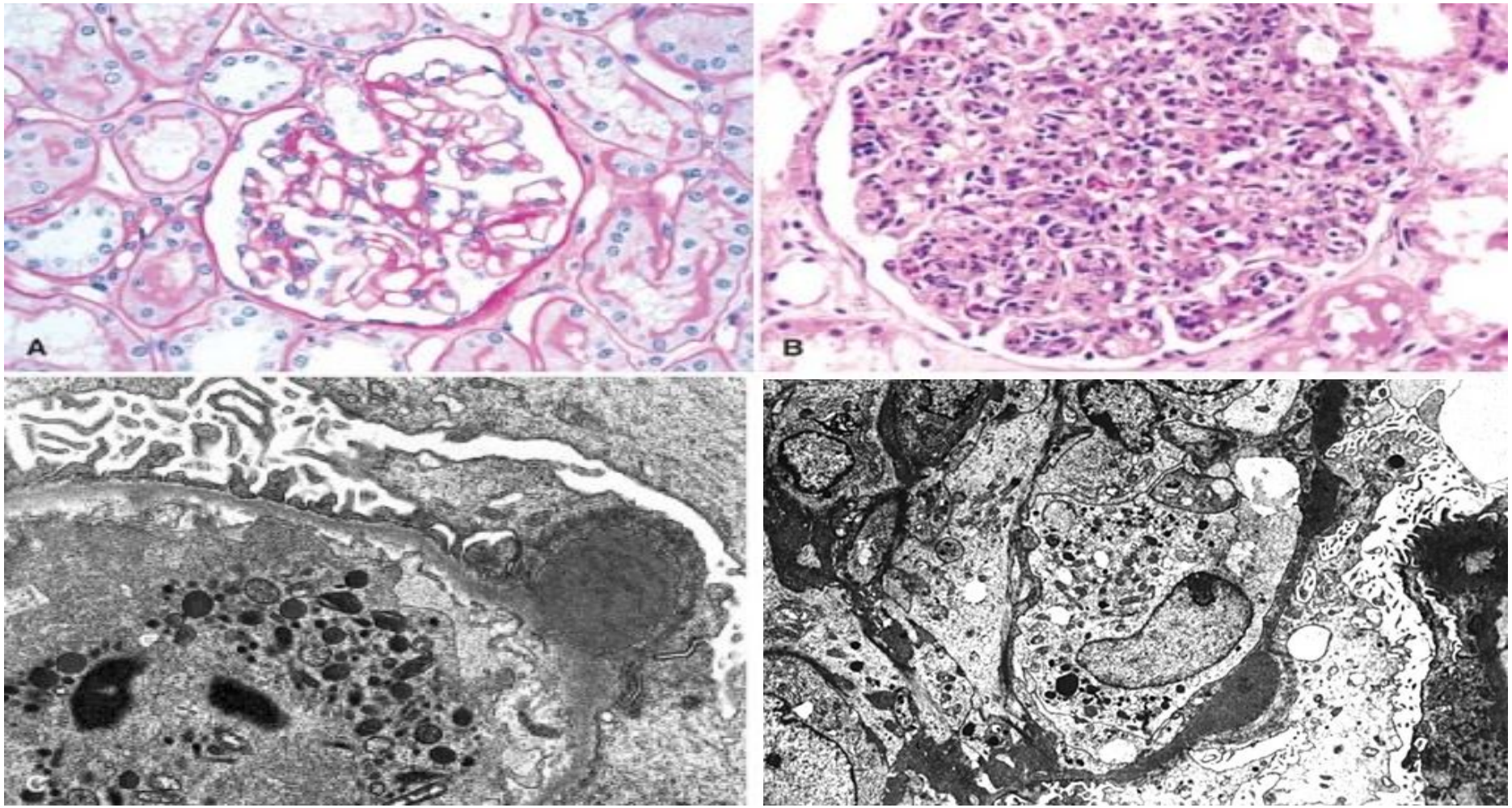
o Group A β-hemolytic streptococci:- Certain strains (1,4,12) are nephritogenic.

o Increased titers of antistreptococcal antibodies:- Anti Streptolysin O (ASO), anti DNAase B, and anti cationic proteinase .

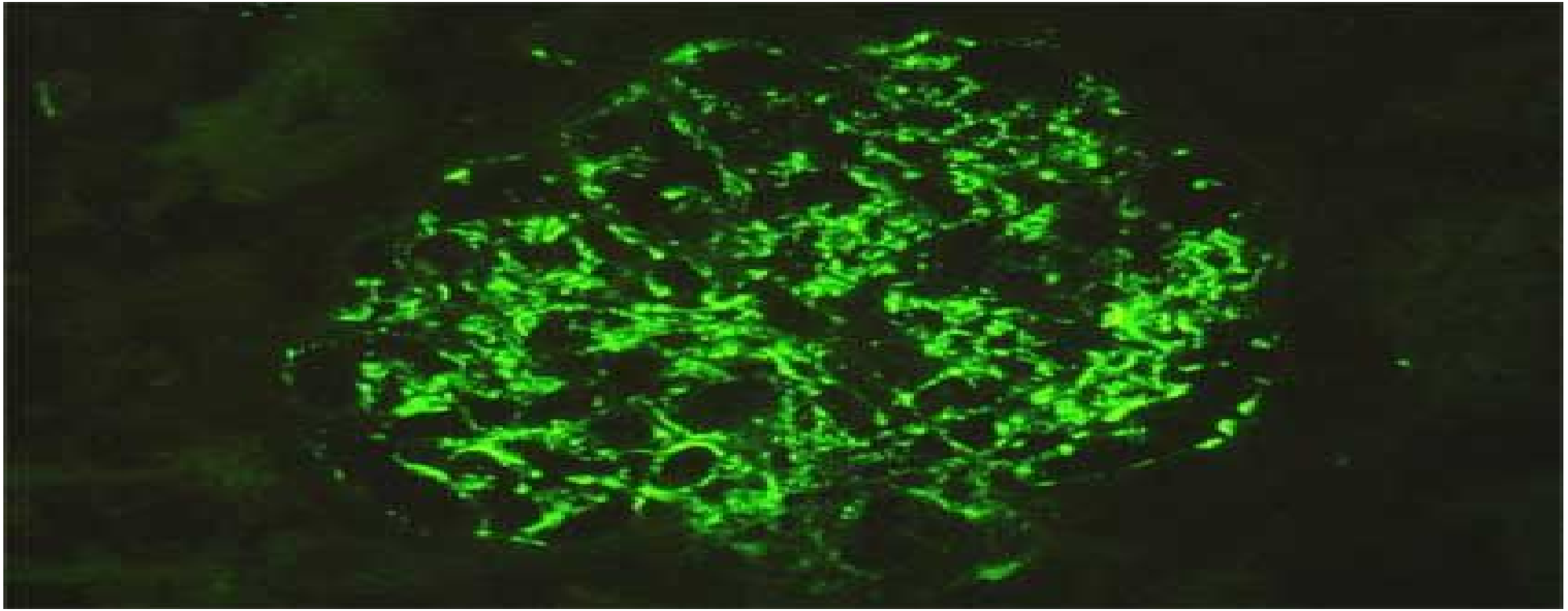
o Granular immune complex deposits in glomeruli.

APGN: A, Normal glomerulus. B, Glomerular hypercellularity is due to intracapillary leukocytes and proliferation of intrinsic glomerular cells.

C, EM: Typical electron-dense subepithelial hump (above GBM and below the Parietal epithelial cell layer), and a neutrophil in capillary lumen.



By immunofluorescence microscopy:
Granular immune complex deposits (of IgG, IgM, and C3) in mesangium and glomerular BM.

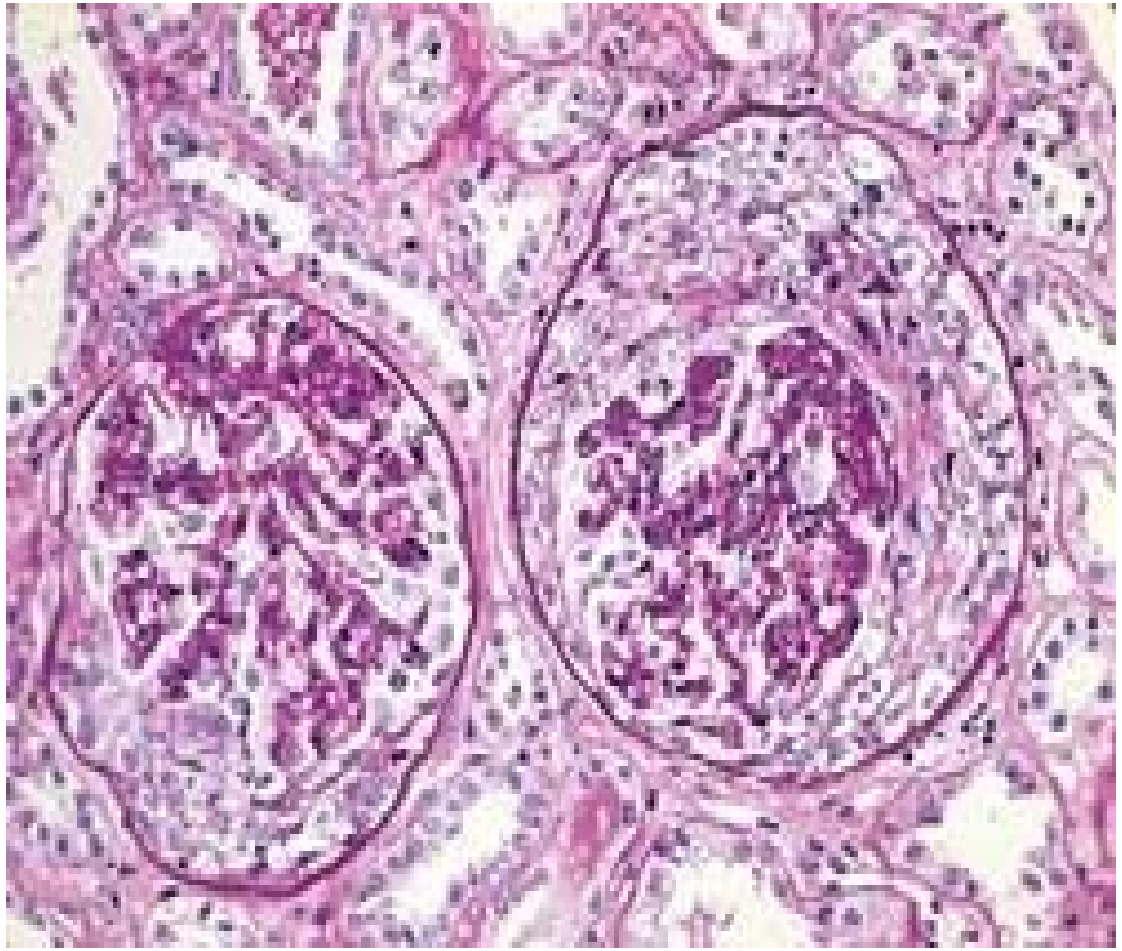


Rapidly progressive (Crescentic) GN:

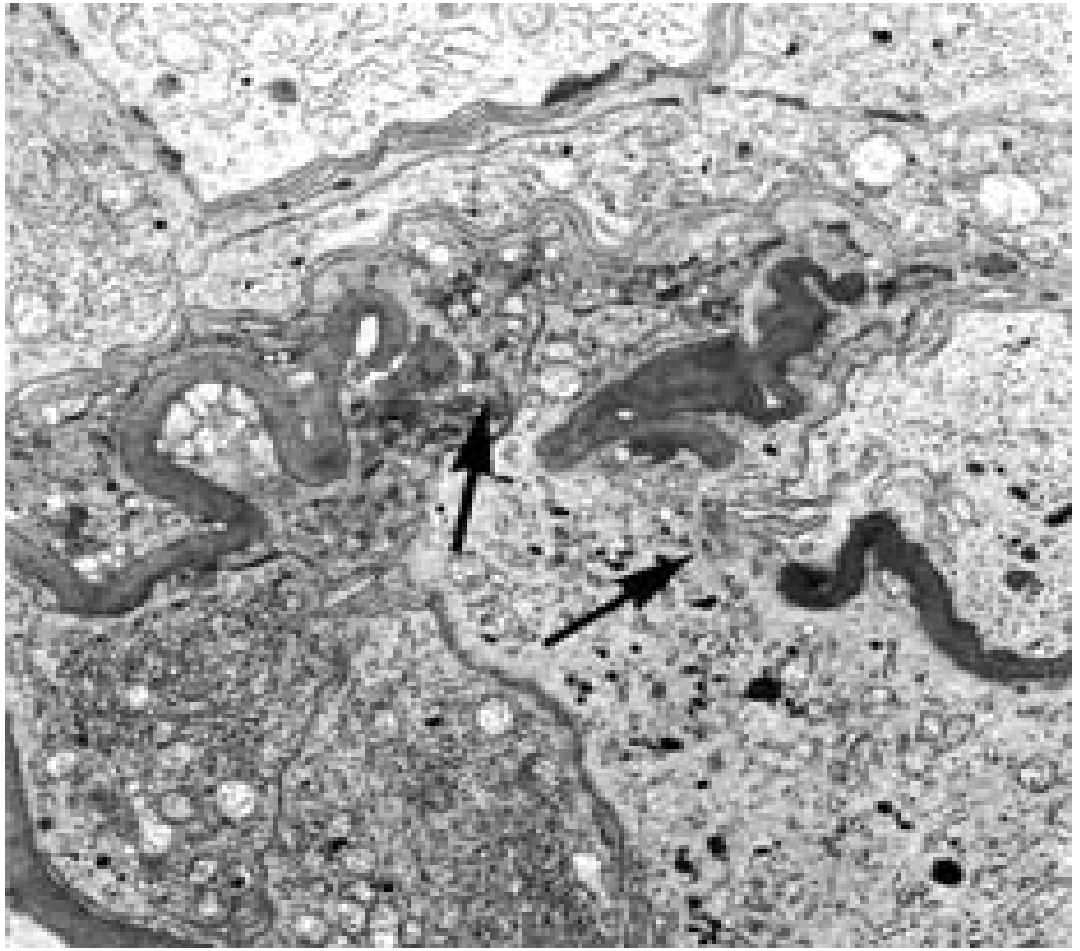
- Rapid and progressive loss of renal function with severe oliguria.
- If untreated death from renal failure within weeks to months.
- The crescents are produced by:- proliferation of parietal epithelial cells, fibrin deposition, and infiltration of leukocytes.
- Classification and Pathogenesis: RPGN divided into three groups on basis of immunologic findings:

  - Type I RPGN (Anti-GBM Antibody): Goodpasture syndrome.
  - Type II RPGN (Immune Complex): Postinfectious.
  - Type III RPGN (Pauci-Immune) [Lack of anti-GBM antibodies or immune complexes]: ANCA associated.

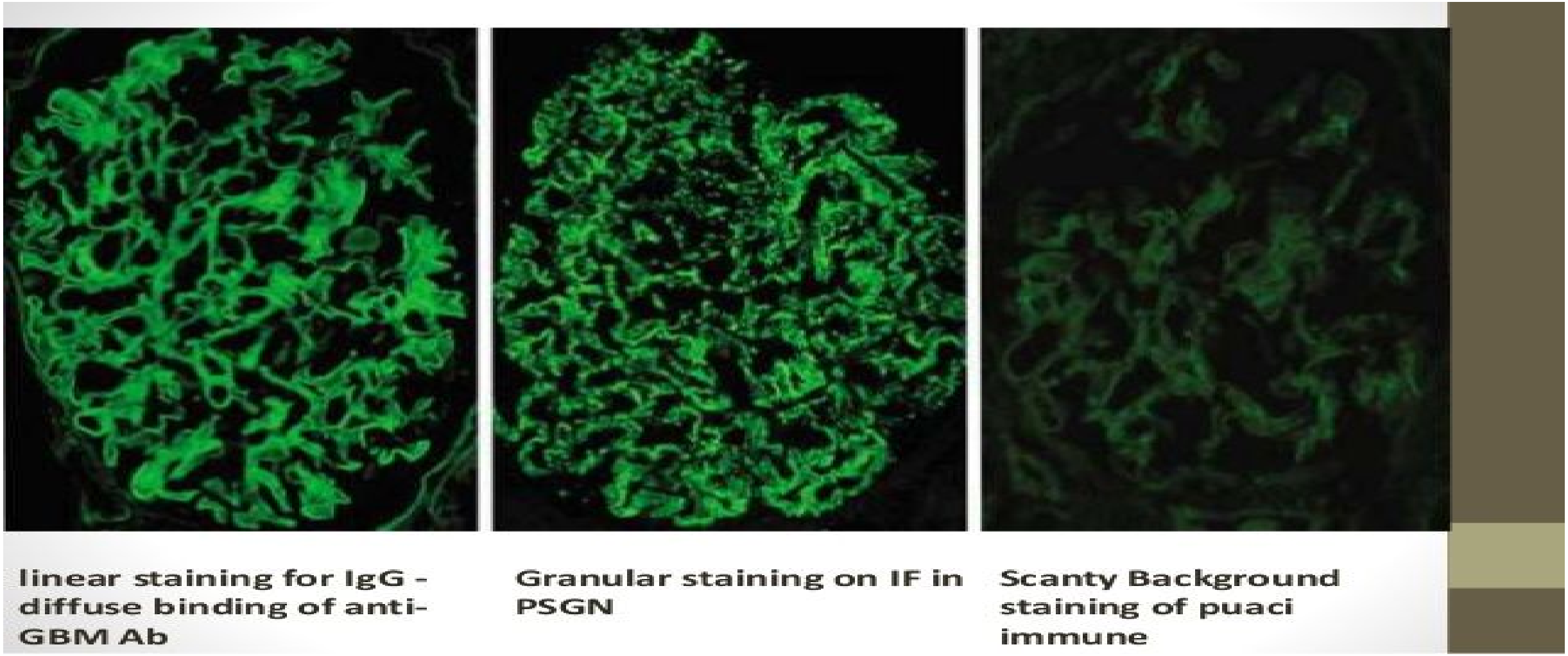
Morphology:



Crescentic GN (PAS stain). Note the collapsed glomerular tufts and the crescent-shaped mass of proliferating cells and leukocytes internal to Bowman capsule.



E.M: characteristic wrinkling of GBM with focal disruptions in its continuity (arrows).



linear staining for IgG - diffuse binding of anti-GBM Ab

Granular staining on IF in PSGN

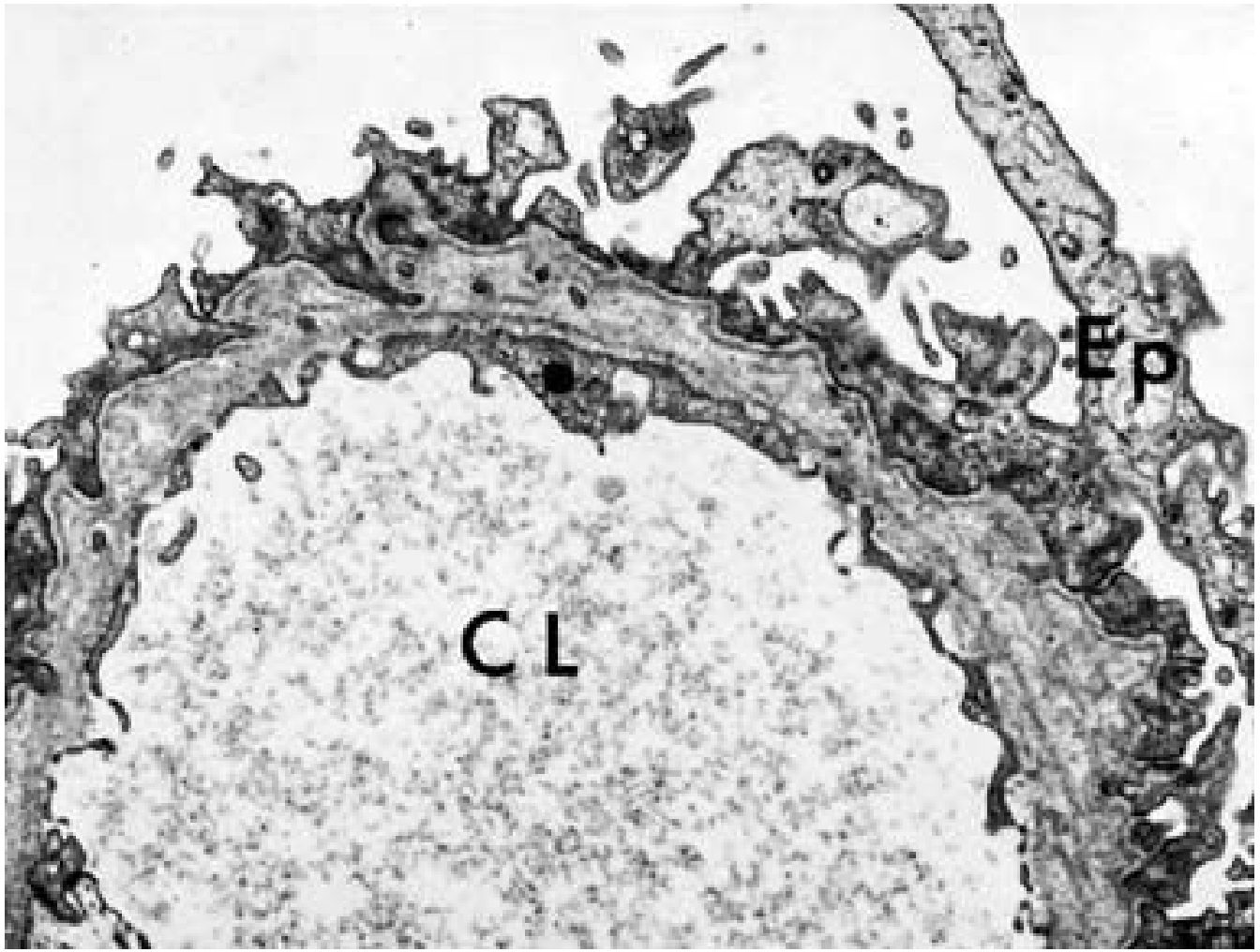
Scanty Background staining of puaci immune

Immunofluorescence microscopy: 1. Goodpasture syndrome cases show linear fluorescence for immunoglobulin and complement, 2. post infectious cases exhibit granular immune deposits; 3. pauci-immune cases have little or no deposition of immune reactants.

Hereditary nephritis:
Alport Syndrome:

- X-linked form: most common, males express full syndrome, and females are carriers.
- AD and AR forms: rare, both males and females express full syndrome.
- When fully developed: nephritis progressing to CRF, nerve deafness, and eye disorders (lens dislocation).
- In carriers: typically limited to hematuria.
- Pathogenesis: Defective GBM synthesis because of mutation in the gene for α5 chain of type IV collagen.

Morphology;
EM: GBM shows irregular foci of thickening alternating with attenuation (thinning) with a basket-weave appearance. Similar alterations can be found in tubular BM.



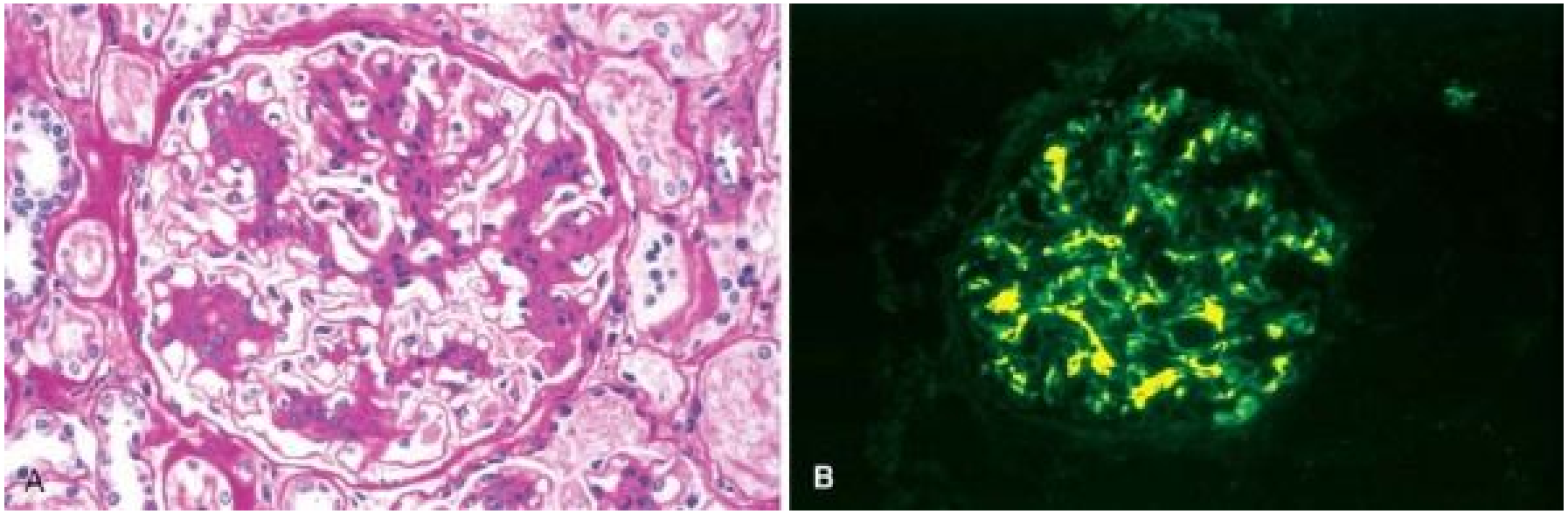
IgA Nephropathy (Berger’s disease):

- Most common type of GN worldwide.
- Can affect all ages but is most common in second and third decades of life.
- Presence of prominent IgA deposits in mesangium.
- Can be suspected by light microscopy, but diagnosis is made only by immunofluorescence.
- Frequent cause of recurrent gross or microscopic hematuria.
- Mild proteinuria is usually present.
- Occasionally develop NS.
- Rarely develop rapidly progressive crescent GN.

Pathogenesis;

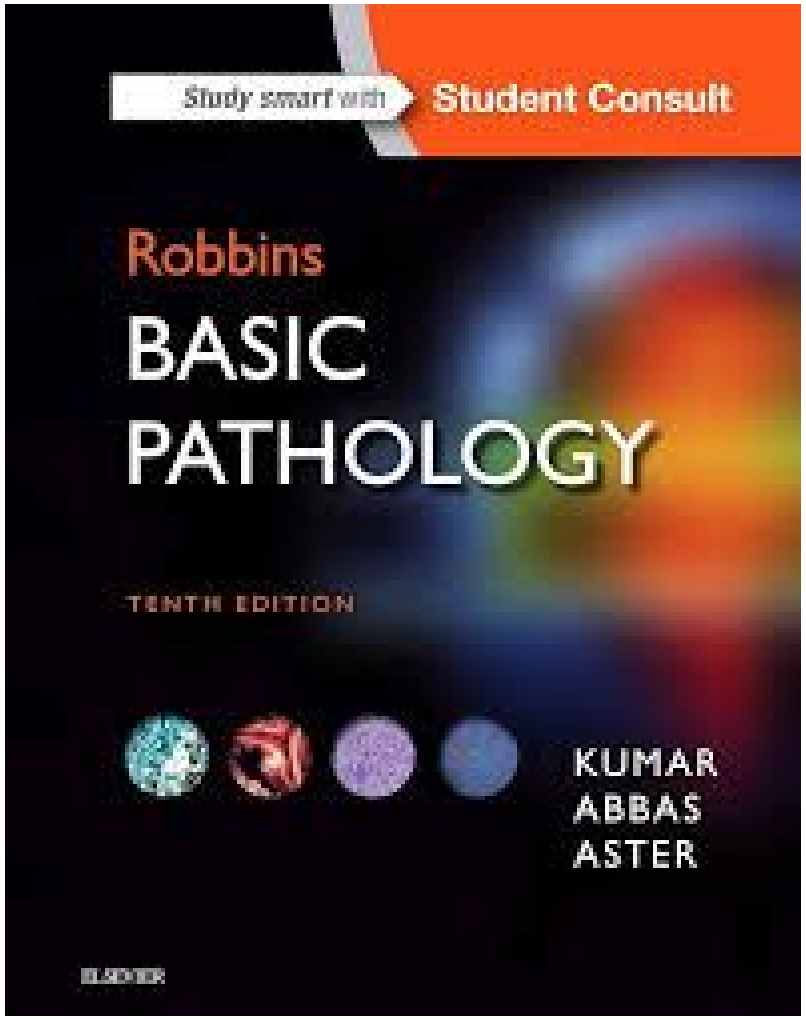
- Genetic or acquired abnormality of immune regulation leading to increased mucosal IgA synthesis in response to respiratory or gastrointestinal exposure to environmental agents (e.g., viruses, bacteria, food proteins).
- IgA containing immune complexes are then trapped in mesangium.

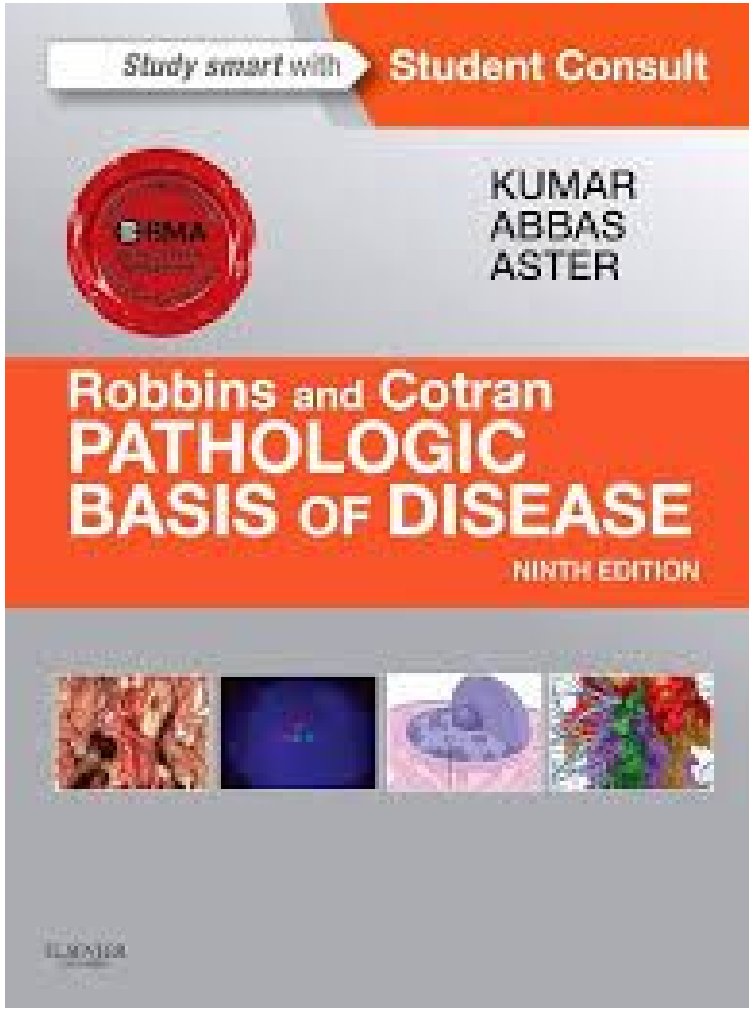
Morphology;



IgA nephropathy: *A*: Light microscopy showing mesangial proliferation and increase mesangial matrix . *B*: immunofluorescence: deposition of IgA, principally in mesangium.

Reference


Study smart with
Student Consult
Robbins
BASIC
PATHOLOGY
TENTH EDITION
KUMAR
ABBAS
ASTER


Study smart with
Student Consult
KUMAR
ABBAS
ASTER
Robbins and Cotran
PATHOLOGIC
BASIS OF DISEASE
NINTH EDITION